# Component-based Software Engineering

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# Topic

- Part 1: Introduction to CBSE
- Part 2: Components in Spring Boot: Foundations
- Part 3: Component Interfaces, Composition, and Lifecycle
- Part 4: Advanced Component Features in Spring Boot
- Part 5: Component-Based Architecture with Spring Boot
- Part 6: Challenges, Best Practices, and Wrap-Up

# Part 1: Introduction to CBSE

- CBSE is the software engineering practices that developing software mainly based on software components.

- We will focus on three core principles: **Components**, **Interfaces**, and **Composition**

- Beyond the core principles, understanding CBSE requires a deeper look at **components**, **component models**, and **middleware**, as well as how they interrelate.

# Benefits and Challenges

**Benefits**

- Reusability reduces development time and costs.
- Modularity enhances maintainability and scalability.
- Easier testing and debugging of isolated components.

**Challenges**

- Dependency management and version conflicts.
- Overhead from inter-component communication.
- Designing components that are both reusable and specific enough for practical use.

# Components

- In CBSE, a component is a modular, reusable, and independently deployable unit of software that encapsulates specific functionality and data.
- It provides services to other parts of the system through well-defined interfaces, hiding its internal implementation details.
- This modularity promotes reusability and maintainability.

# Components

Here are the summarized characteristics of components:

- **Self-contained:** Each component encapsulates its own logic and data.
- **Reusable:** Designed to be used across different systems or contexts.
- **Replaceable:** Can be swapped with another component that provides the same interface.
- **Interface-driven:** Interact with other components solely through defined interfaces, promoting loose coupling and high cohesion.

# Components in Spring Boot

- In Spring Boot, components are typically classes annotated with @Component, @Service, @Repository, or @Controller.

- These annotations tell the Spring Inversion of Control (IoC) container to manage the class, handling its instantiation, dependency injection, and lifecycle

- UserService is a self-contained component that provides a specific service (fetching user data).

- Other parts of the application can use it without knowing how it retrieves the data, aligning with CBSE's emphasis on encapsulation and modularity.

```
@Service
public class UserService {
    public User getUserById(Long id) {
        // Logic to fetch a user by ID
        return new User(id, "John Doe");
    }
}
```

# Interfaces

- Interfaces in CBSE define the contract that components must follow.
- They specify what services a component provides (and sometimes requires) without exposing its internal workings.
- This abstraction enables loose coupling, as components can be swapped out as long as they adhere to the same interface.

# Interfaces in Spring Boot

- In Spring, interfaces are commonly used to define service contracts, with concrete implementations injected at runtime.

- The UserRepository interface acts as a contract, allowing different implementations (e.g., JPA, JDBC, or in-memory) to be used interchangeably.

- This reflects CBSE's focus on well-defined interfaces for component interaction.

```java
public interface UserRepository {
    User findById(Long id);
}

@Repository
public class JpaUserRepository implements UserRepository {
    @Override
    public User findById(Long id) {
        // JPA-specific implementation to fetch user from database
        return new User(id, "Jane Doe");
    }
```

# Composition

- Composition in CBSE involves assembling components into a complete system by connecting them through their interfaces.

- This process leverages dependency relationships to create a functional whole, ensuring that components work together seamlessly.

# Composition in Spring Boot

- Spring Boot uses dependency injection (DI) to compose components.
- The UserService is composed with a UserRepository via constructor inject
- Spring's IoC container manages this composition, connecting the componer at runtime based on their interfaces.

```
@Service
public class UserService {
    private final UserRepository userRepository;

    @Autowired
    public UserService(UserRepository userRepository) {
        this.userRepository = userRepository;
    }

    public User getUserById(Long id) {
        return userRepository.findById(id);
    }
}
```

# Component Models

A component model is a framework or set of standards that governs how components are defined, implemented, and composed.

It provides:

- **Conventions:** Rules for creating components (e.g., how to specify interfaces).
- **Interaction Mechanisms:** How components communicate (e.g., method calls, events).
- **Lifecycle Management:** How components are instantiated, used, and destroyed.

# Component Models

- Examples of component models include
  - Enterprise JavaBeans (EJB)
  - Microsoft’s COM
  - Spring

- A component model ensures consistency and interoperability across a system.

# Spring Component Model

Relies on Plain Old Java Objects (POJOs) enhanced with annotations:

- @Component (and its derivatives) identifies components.

- @Autowired specifies dependencies.

- Spring Boot’s auto-configuration further simplifies this by automatically setting up components based on classpath dependencies.

# Component and Component Models

- The component model defines the “rules of the game” for components.

- Components are instances that conform to these rules, ensuring they can be managed and composed effectively within the system.

# Middleware

Middleware is software that sits between the operating system and applications, providing services that simplify component interaction and system integration.

It handles:

- **Communication:** Enabling components to talk to each other, even across networks (e.g., via RPC or messaging).
- **Data Management:** Abstracting database access or caching.
- **Infrastructure Services:** Offering security, transaction management, or load balancing.

# Middleware

In CBSE, middleware acts as the “glue” that supports component composition and execution, especially in distributed systems.

**Spring Example:**

- **Spring Data:** Acts as middleware by providing a consistent data access layer between components and databases.
- **Spring Cloud:** Offers middleware features like service discovery (Eureka), load balancing (Ribbon), and circuit breaking (Hystrix) for distributed Spring Boot applications.
- For instance, a Spring Boot app might use Spring Cloud to connect a UserService component on one server with a UserRepository component on another, abstracting the network complexity.

# Components, Component Models, and Middleware

- **Components and Component Models:** The component model defines how components are structured and interact. Components are the concrete realizations of this model.

- **Components and Middleware:** Middleware provides the runtime environment and services that components rely on to function. A component like UserService might use middleware (e.g., Spring Data) to access a database without handling the low-level details itself.

# Components, Component Models, and Middleware

- **Component Models and Middleware:** The component model often assumes a specific middleware environment. Spring's model, for instance, integrates tightly with its middleware (e.g., Spring Cloud), ensuring components can leverage distributed system features seamlessly.

Together, they form a layered system:

1. **Components** deliver customized functionality.
2. **Component Models** standardize their creation and composition.
3. **Middleware** enables their execution and interaction, especially in complex or distributed scenarios.

# Part 2: Components in Spring Boot: Foundations

**Objectives**

- Learn how Spring Boot implements CBSE principles.
- Understand component definition and basic dependency injection.

# Spring Boot Overview

Spring Boot is an extension of the Spring Framework that streamlines the development of production-ready applications by reducing configuration complexity and providing sensible defaults.

Key features include:

- **Auto-configuration**: Spring Boot automatically configures components based on the dependencies present in the classpath.
- **Embedded Servers**: Built-in support for web servers like Tomcat or Jetty allows applications to run without external server setup.
- **Opinionated Defaults**: Provides pre-configured settings for common tasks, which developers can override as needed.

# Spring Boot for CBSE

Spring Boot’s architecture is inherently component-based, making it an ideal framework for implementing CBSE principles:

- Components are managed by the Spring IoC container, ensuring modularity and reusability.
- Dependency injection facilitates loose coupling between components.
- Auto-configuration and starter dependencies simplify the integration and composition of components into cohesive applications.

# Components in Spring Boot

A **component** in Spring is any class that is managed by the Spring IoC container. These classes are typically annotated to indicate their role and enable automatic detection by Spring.

- **Core Annotations**:
  - **@Component**: Spring-managed bean.
  - **@Service**: Encapsulates business logic.
  - **@Repository**: Data access component.
  - **@Controller / @RestController**: Components for HTTP requests handling.
- **Component Registration**:
  Spring Boot uses a process called **component scanning** to automatically detect and register components. By default, Spring scans the package containing the main application class and all its sub-packages.

# Components in Spring Boot

- Example

```java
import org.springframework.stereotype.Service;

@Service
public class GreetingService {
    public String greet() {
        return "Hello, World!";
    }
}
```

- The @Service annotation marks GreetingService as a component.
- Spring Boot detects it during component scanning and registers it as a bean in the IoC container.

# Inversion of Control (IoC) Container

The Spring's component management system, embodying the **IoC** principle.

In IoC, the framework takes control of creating and managing objects, rather than the application code doing so manually.

- **Role of the IoC Container**:
  - Instantiates components (referred to as **beans**).
  - Configures them by injecting dependencies and setting properties.
  - Manages their lifecycle from creation to destruction.
  - Stores beans in the **application context**, which serves as the runtime environment.

# Dependency Injection (DI)

- **Dependency Injection** is a design pattern where a component's dependencies are provided by the framework rather than being instantiated within the component itself.

- This enhances modularity, testability, and maintainability.

- **Types of Dependency Injection**:
  - **Constructor Injection**: Recommended for mandatory dependencies.
  - **Setter Injection**: Suitable for optional dependencies or runtime changes.
  - **Field Injection**: Direct injection into fields (less preferred due to testability concerns)

# Constructor Injection

```
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

@Service
public class OrderService {
    private final PaymentService paymentService;

    @Autowired
    public OrderService(PaymentService paymentService) {
        this.paymentService = paymentService;
    }
}
```

# Setter Injection

```
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

@Service
public class NotificationService {
    private EmailService emailService;

    @Autowired
    public void setEmailService(EmailService emailService) {
        this.emailService = emailService;
    }
}
```

# Field Injection

```
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

@Service
public class UserService {
    @Autowired
    private UserRepository userRepository;
}
```

# Advantages of DI

- **Advantages of DI**:
  - **Decoupling**: Components are independent of how their dependencies are created.
  - **Testability**: Dependencies can be mocked or stubbed in tests.
  - **Flexibility**: Dependencies can be swapped or reconfigured without altering component code.

# Component Discovery and Customization

- Spring Boot’s components are discovered during component scanning, but it also offers customization options for fine-grained control.
- **Component Scanning**: By default, Spring scans the package of the main application class and its sub-packages.
- Customize this behavior with @ComponentScan:

```
@SpringBootApplication
@ComponentScan(basePackages = {"com.example.services", "com.example.repositories"}
public class MyApplication {
    public static void main(String[] args) {
        SpringApplication.run(MyApplication.class, args);
    }
}
```

# Component Discovery and Customization

- Define beans explicitly in a configuration class using @Bean
- This approach is useful for third-party classes or when more control is needed

```
import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import com.zaxxer.hikari.HikariDataSource;

@Configuration
public class AppConfig {
    @Bean
    public DataSource dataSource() {
        return new HikariDataSource();
    }
}
```

# Practical Example: Building a Simple REST API

```
public class Product {
    private Long id;
    private String name;
    private double price;

    public Product(Long id, String name, double price) {
        this.id = id;
        this.name = name;
        this.price = price;
    }

    // Getters and setters
    public Long getId() { return id; }
    public void setId(Long id) { this.id = id; }
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
    public double getPrice() { return price; }
    public void setPrice(double price) { this.price = price; }
}
```

# Practical Example: Building a Simple REST API

```java
import org.springframework.stereotype.Service;
import java.util.Arrays;
import java.util.List;

@Service
public class ProductService {
    public List<Product> getAllProducts() {
        // Hardcoded for simplicity
        return Arrays.asList(
            new Product(1L, "Laptop", 999.99),
            new Product(2L, "Smartphone", 499.99)
        );
    }
}
```

# Practical Example: Building a Simple REST API

```java
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;
import java.util.List;

@RestController
@RequestMapping("/api/products")
public class ProductController {
    private final ProductService productService;

    @Autowired
    public ProductController(ProductService productService) {
        this.productService = productService;
    }

    @GetMapping
    public List<Product> getProducts() {
        return productService.getAllProducts();
    }
```

# Practical Example: Building a Simple REST API

```
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class ProductApplication {
    public static void main(String[] args) {
        SpringApplication.run(ProductApplication.class, args);
    }
}
```

# Part 3: Component Interfaces, Composition, and Lifecycle

**Objectives**

- Understand the use of interfaces for component abstraction.
- Explore component composition techniques and lifecycle management.

# Interfaces in CBSE and Spring Boot

Interfaces are a cornerstone of CBSE because they establish a **contract** that components must follow.

This contract specifies what services a component offers (its capabilities) and what it requires (its dependencies).

# Interfaces in CBSE and Spring Boot

By defining interactions through interfaces rather than concrete implementations, interfaces enable:

- **Polymorphism**: Different implementations of the same interface can be swapped seamlessly.
- **Loose Coupling**: Components depend on abstractions, reducing tight dependencies on specific implementations.
- **Testability**: Interfaces simplify mocking dependencies during unit testing.

# Spring Boot Example: Defining a Service Interface

- The PaymentProcessor interface defines a contract for processing payments.
- CreditCardProcessor and PayPal Processor provide concrete implementations that can be used interchangeably.

```
public interface PaymentProcessor {
    void processPayment(double amount);
}

@Service
public class CreditCardProcessor implements PaymentProcessor {
    @Override
    public void processPayment(double amount) {
        System.out.println("Processing credit card payment of $" + amount);
    }
}

@Service
public class PayPalProcessor implements PaymentProcessor {
    @Override
    public void processPayment(double amount) {
        System.out.println("Processing PayPal payment of $" + amount);
    }
}
```

# Benefits of Using Interfaces

- **Flexibility**: Switch between payment processors (e.g., credit card to PayPal) by injecting a different implementation.
- **Extensibility**: Add new payment methods (e.g., CryptoProcessor) without altering existing code.
- **Testability**: Mock the PaymentProcessor interface in tests to isolate dependent components.

# Composition Techniques

- In CBSE, **composition** refers to assembling components into a functional system by connecting them through their interfaces.
- Spring Boot achieves this primarily through **dependency injection (DI)**, a mechanism where the framework provides a component's dependencies automatically.
- Spring supports several DI techniques, each suited to different scenarios.

# Best Practice for Dependency Injection

- **Prefer Constructor Injection:** Use it for mandatory dependencies to ensure immutability and explicitness.
- **Use Setter Injection Sparingly:** Reserve it for optional or reconfigurable dependencies.
- **Best Practice for Dependency Injection** Opt for constructor or setter injection to improve testability and maintainability.

# Component Lifecycle Management

- Spring Boot’s Inversion of Control (IoC) container manages the entire lifecycle of components (beans), from instantiation to destruction.

- Understanding this lifecycle allows developers to perform initialization tasks (e.g., loading resources) and cleanup tasks (e.g., closing connections) at the appropriate stages.

# Key Phases of the Component Lifecycle

- **Creation**
  - The IoC container creates the bean instance.
  - Dependencies are injected based on the chosen DI method (constructor, setter, or field).
- **Initialization**
  - After dependency injection, Spring invokes initialization logic.
  - **@PostConstruct**: Annotate a method to run it after the bean is fully constructed.
- **Usage**
  - The bean is fully initialized and available for use by other components or services.
- **Destruction**
  - Before the application context closes, Spring invokes destruction logic.
  - **@PreDestroy**: Annotate a method to run it before the bean is destroyed.

```java
@Service
public class CacheService {
    private Map<String, Object> cache = new HashMap<>();

    @PostConstruct
    public void init() {
        System.out.println("CacheService initialized. Loading cache...");
        // Simulate loading initial cache data
        cache.put("defaultKey", "defaultValue");
    }

    public void put(String key, Object value) {
        cache.put(key, value);
    }

    public Object get(String key) {
        return cache.get(key);
    }

    @PreDestroy
    public void cleanup() {
        System.out.println("CacheService shutting down. Clearing cache...");
        cache.clear();
    }
}
```

# Why Lifecycle Management Matters

- **Resource Initialization**: Set up connections, load configurations, or pre-populate data (e.g., cache).
- **Resource Cleanup**: Release resources like database connections, file handles, or memory to prevent leaks.
- **Controlled Behavior**: Ensure components start and stop gracefully, maintaining application stability.

# Best Practices for Component Design

- Define Interfaces for Abstraction
- Favor Constructor Injection
- Single Responsibility Principle
- Leverage Lifecycle Hooks (@PostConstruct and @PreDestroy)
- Avoid Circular Dependencies
- Document Interfaces (e.g., JavaDoc)

# Practical Example: Composing Components with Interfaces

**Scenario**

- An OrderService depends on a PaymentProcessor (interface with multiple implementations) and a NotificationService. The system processes payments and sends notifications when an order is placed.

1. Payment processor interface

```
public interface PaymentProcessor {
    void processPayment(double amount);
}
```

# Practical Example: Composing Components with Interface

- Implement concrete payment processors

```
@Service
public class CreditCardProcessor implements PaymentProcessor {
    @Override
    public void processPayment(double amount) {
        System.out.println("Processing credit card payment of $" + amount);
    }
}

@Service
public class PayPalProcessor implements PaymentProcessor {
    @Override
    public void processPayment(double amount) {
        System.out.println("Processing PayPal payment of $" + amount);
    }
}
```

# Practical Example: Composing Components with Interface

- Define NotificationService

```
@Service
public class NotificationService {
    public void sendNotification(String message) {
        System.out.println("Sending notification: " + message);
    }
}
```

# Practical Example: Composing Components with Interface

- Compose the OrderService with dependencies

```
@Service
public class OrderService {
    private final PaymentProcessor paymentProcessor;
    private final NotificationService notificationService;

    @Autowired
    public OrderService(PaymentProcessor paymentProcessor, NotificationService
notificationService) {
        this.paymentProcessor = paymentProcessor;
        this.notificationService = notificationService;
    }

    public void placeOrder(double amount, String customerEmail) {
        paymentProcessor.processPayment(amount);
        notificationService.sendNotification("Order placed successfully for " +
customerEmail);
    }
}
```

# Practical Example: Composing Components with Interface

- Resolve multiple implementations
- Option 1: @Primary

```
@Service
@Primary
public class CreditCardProcessor implements PaymentProcessor {
    @Override
    public void processPayment(double amount) {
        System.out.println("Processing credit card payment of $" + amount);
    }
}
```

# Practical Example: Composing Components with Interface

- Resolve multiple implementations
- Option 2: @Qualifier

```
@Service
public class OrderService {
    private final PaymentProcessor paymentProcessor;
    private final NotificationService notificationService;

    @Autowired
    public OrderService(@Qualifier("creditCardProcessor") PaymentProcessor
paymentProcessor,
                        NotificationService notificationService) {
        this.paymentProcessor = paymentProcessor;
        this.notificationService = notificationService;
    }

    // ... rest of the code
}
```

# Part 4: Advanced Component Features in Spring Boot

- **Objective**:
  - Explore advanced Spring Boot component management techniques, including custom scopes, lazy loading, dynamic dependency injection, and conditional bean creation
  - Design flexible and scalable applications aligned with Component-Based Software Engineering (CBSE) principles.

- **Prerequisites**: Complete hands-on exercises 1 and 2

# Overview

This section introduces advanced Spring Boot features that enhance component flexibility and adaptability:

- **Component Scopes with Real-World Scenarios**
- **Advanced Dependency Resolution**
- **Programmatic Bean Definition with Conditions**

These concepts enable you to tackle real-world challenges like circular dependencies, runtime configuration, and dynamic component selection, reinforcing CBSE’s emphasis on modularity and reusability.

# Component Scopes with Real-World Scenarios

Spring Boot manages component lifecycles through **scopes**, determining how instances are created and shared. Advanced scopes and custom implementations offer fine-grained control.

- **Standard Scopes**:
  - **Singleton**: One instance per ApplicationContext (default).
  - **Prototype**: New instance per request.
- **Web-Specific Scopes**:
  - **Request**: One instance per HTTP request.
  - **Session**: One instance per user session.
  - **Application**: One instance per ServletContext (web app lifecycle).
- **Custom Scopes**: Define bespoke scopes for specific use cases (e.g., thread-local or transaction-specific).

# Example: Request and Session Scopes

- Check the *sample code* to see how @RequestScope and @SessionScope are used

Request Scopes

- Practical Scenario 1: Tracking Request-Specific Metrics
  - In an e-commerce API, you want to track metrics for each incoming request (e.g., processing time, number of database queries) without persisting this data beyond the request.
- Practical Scenario 2: Temporary User Input Validation
  - A user submits a complex form (e.g., product customization options) via a POST request, and you need to validate and process it within a single request.

# Example: Request and Session Scopes

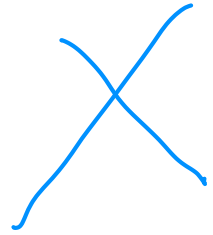
Session scopes

- Practical Scenario 1: Shopping Cart Management
  - In the e-commerce API, users add items to a shopping cart across multiple requests before checking out.
- Practical Scenario 2: Multi-Step User Registration Wizard
  - A user completes a multi-step registration process (e.g., personal info, shipping address, payment details) across several requests.

# Advanced Dependency Resolution 高级依赖解析

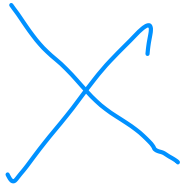
- Complex applications require sophisticated dependency management.
- Real-world systems often involve intricate relationships between components, dynamic runtime conditions, and the need for flexibility, scalability, and maintainability.
- As applications grow in complexity advanced techniques like @Lazy, ObjectProvider, @Qualifier, and programmatic bean definitions are necessary.

# Why Sophisticated Dependency Management is Needed



- **Ambiguity in Dependency Resolution**: When multiple beans implement the same interface
- **Circular Dependencies**: In tightly coupled designs, two beans may depend on each other, causing Spring to fail during context initialization.
- **Dynamic or Conditional Dependencies**: Applications may need to select dependencies at runtime based on configuration, environment, or user input
- **Performance Optimization**: Eager initialization of all beans can slow startup in large applications

# Why Sophisticated Dependency Management is Needed



- **Scalability and Flexibility**: As systems scale (e.g., microservices or modular designs), hardcoding dependencies with @Autowired limits adaptability to new requirements or implementations
- **Testing and Maintenance**: Basic @Autowired tightly couples components, making it harder to mock or replace dependencies in tests or during maintenance.

# Ambiguity in Dependency Resolution

- When multiple beans implement the same interface, we use @Primary or @Qualifier to specify the required implementation.

- Now let explore how ObjectProvider enhance more flexibility

```java
public interface PaymentProcessor {
    void process(double amount);
}

@Service
@Qualifier("creditCard")
public class CreditCardProcessor implements PaymentProcessor {
    @Override
    public void process(double amount) {
        System.out.println("Processing $" + amount + " via Credit Card");
    }
}

@Service
@Qualifier("paypal")
public class PayPalProcessor implements PaymentProcessor {
    @Override
    public void process(double amount) {
        System.out.println("Processing $" + amount + " via PayPal");
    }
}
```

```java
@Service
public class DynamicOrderService {
    private final ObjectProvider<PaymentProcessor> processors;

    public DynamicOrderService(ObjectProvider<PaymentProcessor> processors) {
        this.processors = processors;
    }

    public void placeOrder(double amount, String paymentType) {
        PaymentProcessor processor = processors.stream()
            .filter(p ->
p.getClass().getSimpleName().toLowerCase().contains(paymentType.toLowerCase()))
            .findFirst()
            .orElseThrow(() -> new IllegalArgumentException("No processor for " +
paymentType));
        processor.process(amount);
    }
}
```

```java
@RestController
@RequestMapping("/orders")
public class OrderController {
    private final DynamicOrderService orderService;

    public OrderController(DynamicOrderService orderService) {
        this.orderService = orderService;
    }

    @PostMapping
    public String createOrder(@RequestParam double amount, @RequestParam String
paymentType) {
        orderService.placeOrder(amount, paymentType);
        return "Order placed";
    }
}
```

# Circular Dependencies

- In the e-commerce API, ProductService needs InventoryService to check stock, and InventoryService needs ProductService to fetch product details for stock updates.

```java
@Service
public class ProductService {
    private final Lazy<InventoryService> inventoryService;

    @Autowired
    public ProductService(Lazy<InventoryService> inventoryService) {
        this.inventoryService = inventoryService;
    }

    public Product getProduct(Long id) {
        Product product = new Product(id, "Laptop", 999.99, 10); // Dummy data
        int stock = inventoryService.get().checkStock(id);
        product.setStock(stock);
        return product;
    }
}
```

```java
@Service
public class InventoryService {
    private final ProductService productService;

    @Autowired
    public InventoryService(ProductService productService) {
        this.productService = productService;
    }

    public int checkStock(Long productId) {
        Product product = productService.getProduct(productId);
        return product.getStock(); // Simplified logic
    }
}
```

# Circular Dependencies

- When you mark a dependency as lazy, Spring does not inject the actual target bean during the startup phase. Instead, it injects a proxy object.

| Step | Action | ProductService | InventoryService |
|---|---|---|---|
| 1 | Spring starts creating ProductService | Instantiating… | Not Started |
| 2 | Spring sees ProductService needs @Lazy InventoryService. Spring creates a proxy and hands it to ProductService. | Created | Not Started |
| 3 | Spring starts creating InventoryService | Created | Instantiating… |
| 4 | Spring sees InventoryService needs ProductService. Spring finds the created ProductService in cache and injects it to InventoryService. | Created | Created |

# Circular Dependencies

- Compare to the use of InventoryService without @Lazy

| Step | Action | Status |
|---|---|---|
| 1 | Spring try to initiate ProductService | Pending… |
| 2 | It sees ProductService needs InventoryService. It pauses ProductService and looks for InventoryService | Searching… |
| 3 | Spring try to initiate InventoryService | Pending… |
| 4 | It sees InventoryService needs ProductService. It pauses InventoryService and looks for ProductService | Searching… |
| 5 | Spring realizes ProductService is still ”In Creation” | Crash |

# Programmatic Bean Definition with Conditions

**Programmatic bean definition** allows dynamic control over component creation, enhanced by conditions and lifecycle customization.

**Programmatic bean definition** in Spring Boot refers to the explicit creation and configuration of beans using Java code, typically via @Configuration classes and @Bean annotations

- **@Configuration and @Bean**: Define beans explicitly.
- **@Conditional**: Create beans based on runtime conditions.
- **BeanPostProcessor**: Customize bean creation globally.

# Why We Need Programmatic Bean Definition

- **Fine-Grained Control Over Bean Creation**: Automatic scanning creates beans based on annotations with limited customization options.
  - **Example**: Configuring a third-party object with specific settings.
- **Dynamic or Conditional Bean Creation**: Beans may need to be created based on runtime conditions
  - **Example**: Loading different implementations based on a property file.
- **Integration with Third-Party Libraries**: Many third-party libraries don’t come with Spring annotations, so you need to wrap them in Spring-managed beans manually.
  - **Example**: Integrating a database connection pool like HikariCP.

# Example: Conditional Bean Definition

- **Scenario**: Enable a logging service only in production

```
@Configuration
public class AppConfig {
    @Bean
    @ConditionalOnProperty(name = "app.environment", havingValue = "prod")
    public LoggingService loggingService() {
        return new LoggingService();
    }
}

public class LoggingService {
    @PostConstruct
    public void init() {
        System.out.println("LoggingService initialized");
    }
}
```

# Example: Configuring a Third-Party Library Bean

- In the e-commerce API, you need to integrate a third-party library (e.g., a payment gateway SDK like Stripe) that doesn’t use Spring annotations.

```
@Configuration
public class PaymentConfig {
    @Bean
    public RequestOptions stripeRequestOptions() {
        Stripe.apiKey = "sk_test_your_key"; // External config
properties)
        return RequestOptions.builder()
                .setConnectTimeout(30_000) // Custom timeout
                .setReadTimeout(80_000)
                .build();
    }
}
```

```
@Service
public class PaymentService {
    private final RequestOptions stripeOptions;

    public PaymentService(RequestOptions stripeOptions) {
        this.stripeOptions = stripeOptions;
    }

    public void processPayment(double amount) {
        System.out.println("Processing $" + amount + " with Stripe options: " +
stripeOptions);
    }
}
```

# Part 5: Component-Based Architecture with Spring Boot

- **Objective**:
  - Explore advanced architectural patterns and techniques in Spring Boot, including Domain-Driven Design (DDD), and hexagonal (ports and adapters) architecture to create a robust, modular e-commerce system aligned with Component-Based Software Engineering (CBSE) principles.

# Overview

This section equips you with the tools to architect complex applications:

- **Modular Design with DDD**: Structure the application using entities, aggregates, and bounded contexts.
- **Hexagonal Architecture**: Implement a ports and adapters pattern for flexibility and testability.

# Modular Design with Domain-Driven Design (DDD)

DDD is a design approach that aligns software structure with business domains, promoting modularity and clarity. Key concepts include:

- **Entities**: Objects with identity (e.g., Product).
- **Aggregates**: Clusters of entities with a root (e.g., Order with OrderItems).
- **Bounded Contexts**: Separate domains with clear boundaries (e.g., Product vs. Order).

# Entity

- An **entity** is an object in the domain model that has a distinct identity and a lifecycle
- entities are defined by a unique identifier (e.g., an ID) that persists across changes to their state
- Entities model the “things” in the domain that need to be tracked individually
- They are the building blocks of the domain model, ensuring that the system reflects real-world concepts with unique identities.

```java
@Data
public class Product {
    private Long id;          // Unique identifier
    private String name;
    private double price;
    private int stock;

    public Product(Long id, String name, double price, int stock) {
        this.id = id;
        this.name = name;
        this.price = price;
        this.stock = stock;
    }

    // Business logic
    public void reduceStock(int quantity) {
        if (quantity > stock) {
            throw new IllegalStateException("Insufficient stock for product " +
id);
        }
        stock -= quantity;
    }

    public void updatePrice(double newPrice) {
        if (newPrice < 0) {
            throw new IllegalArgumentException("Price cannot be negative");
        }
        this.price = newPrice;
```

# Aggregate 聚合

- An **aggregate** is a cluster of related entities and value objects that are treated as a single unit for data consistency and transactional boundaries.
- Each aggregate has a root entity, called the **aggregate root**, which serves as the entry point for interacting with the aggregate.
- Order Aggregate in E-Commerce
  - An Order aggregate includes the Order entity (root) and a collection of OrderItem entities. The total amount must reflect the items, and stock must be reserved atomically.

# Order aggregate

```java
@Data
@Entity
public class Order {
    @Id
    private Long id;                    // Aggregate root identity
    @OneToMany(cascade = CascadeType.ALL, mappedBy = "order")
    private List<OrderItem> orderItems = new ArrayList<>();
    private double totalAmount;
    private String status;

    public Order() {}

    public void addItem(Long productId, int quantity, double unitPrice) {
        OrderItem item = new OrderItem(productId, quantity, unitPrice, this);
        orderItems.add(item);
        calculateTotal();
    }

    // Invariant enforcement
    public void calculateTotal() {
        totalAmount = orderItems.stream()
            .mapToDouble(item -> item.getQuantity() * item.getUnitPrice())
            .sum();
    }
```

```java
@Data
@Entity
public class OrderItem {
    @Id
    @GeneratedValue
    private Long id;
    private Long productId;
    private int quantity;
    private double unitPrice;
    @ManyToOne
    @JoinColumn(name = "order_id")
    private Order order;

    public OrderItem() {}

    public OrderItem(Long productId, int quantity, double unitPrice, Order order) {
        this.productId = productId;
        this.quantity = quantity;
        this.unitPrice = unitPrice;
        this.order = order;
    }
}
```

# Bounded Context

- A **bounded context** is a specific boundary within which a domain model is defined and applicable.
- It’s a way to divide a large, complex domain into smaller, manageable parts, each with its own model, language, and rules.
- Bounded contexts manage complexity by splitting a large domain into focused subdomains, each with a clear purpose.
- ProductCatalog and OrderManagement Bounded Contexts
  - In the e-commerce API, ProductCatalog manages product listings, while OrderManagement handles order processing.

# Bounded Context

Each bounded context with its own:

- **Domain Model**: A set of entities, aggregates, and value objects tailored to the context.
- **Ubiquitous Language**: A shared vocabulary used by developers, domain experts, and the code itself within that context.
- **Rules and Assumptions**: Context-specific business logic and constraints.

- Bounded contexts prevent the domain model from becoming a monolithic, overly generalized mess by allowing different parts of the system to have distinct interpretations of similar concepts (e.g., “Product”) without conflict.

# Bounded Contexts in the E-Commerce Example - ProductCatalog

- The ProductCatalog bounded context is responsible for managing the inventory of products available for sale.

- **Responsibilities**:
  - Maintaining product details (e.g., name, price, stock).
  - Managing stock levels (e.g., reducing stock when reserved).
  - Providing product information to other contexts (e.g., for ordering).

# Bounded Contexts in the E-Commerce Example - ProductCatalog

**Ubiquitous Language**

- **Product**: An item in the inventory with stock available for sale.
- **Stock**: The quantity of a product available for purchase.
- **Reduce Stock**: Reserve stock for an order, ensuring availability.
- **Restock**: Add more units to the inventory from a supplier.

**Boundaries**

- **What's Included**: Product definitions, stock management, and catalog-related operations.
- **What's Excluded**: Order processing, customer details, payment handling.

# Bounded Contexts in the E-Commerce Example - OrderManagement

- The OrderManagement bounded context handles the creation, processing, and tracking of customer orders.

- **Responsibilities**:
  - Creating and managing orders.
  - Calculating order totals based on product prices at the time of ordering.
  - Tracking order status (e.g., pending, confirmed, shipped).

# Bounded Contexts in the E-Commerce Example - OrderManagement

**Ubiquitous Language**

- **Order**: A customer's request to purchase products.
- **Order Item**: A line item in an order, tied to a product and quantity.
- **Total Amount**: The calculated cost of the order based on ordered items.
- **Confirm Order**: Finalize the order after validation.

**Boundaries**

- **What's Included**: Order creation, total calculation, status tracking.
- **What's Excluded**: Stock management, product catalog updates.

# Benefit of Bounded Context

- **Clarity**: “Product” has a clear meaning in each context, avoiding confusion (e.g., stock in ProductCatalog vs. price snapshot in OrderManagement).
- **Modularity**: Each context can evolve independently (e.g., adding supplier logic to ProductCatalog doesn’t affect OrderManagement).
- **Reduced Coupling**: No shared database or model; contexts communicate via well-defined interfaces or events.
- **Scalability**: ProductCatalog and OrderManagement can be deployed separately, aligning with microservices

# Challenges of Bounded Context

- **Context Mapping**: Define how ProductCatalog and OrderManagement relate (e.g., client/supplier relationship where OrderManagement depends on ProductCatalog).
  - Other possible mapping pattern: Anticorruption Layer, Published Language with Event-Driven Integration
- **Duplication**: Some data (e.g., product name) might be duplicated (e.g., in OrderItem for display), but this is intentional to keep contexts independent.
- **Integration Overhead**: REST calls or message queues add complexity (e.g., latency, error handling).

六边形架构

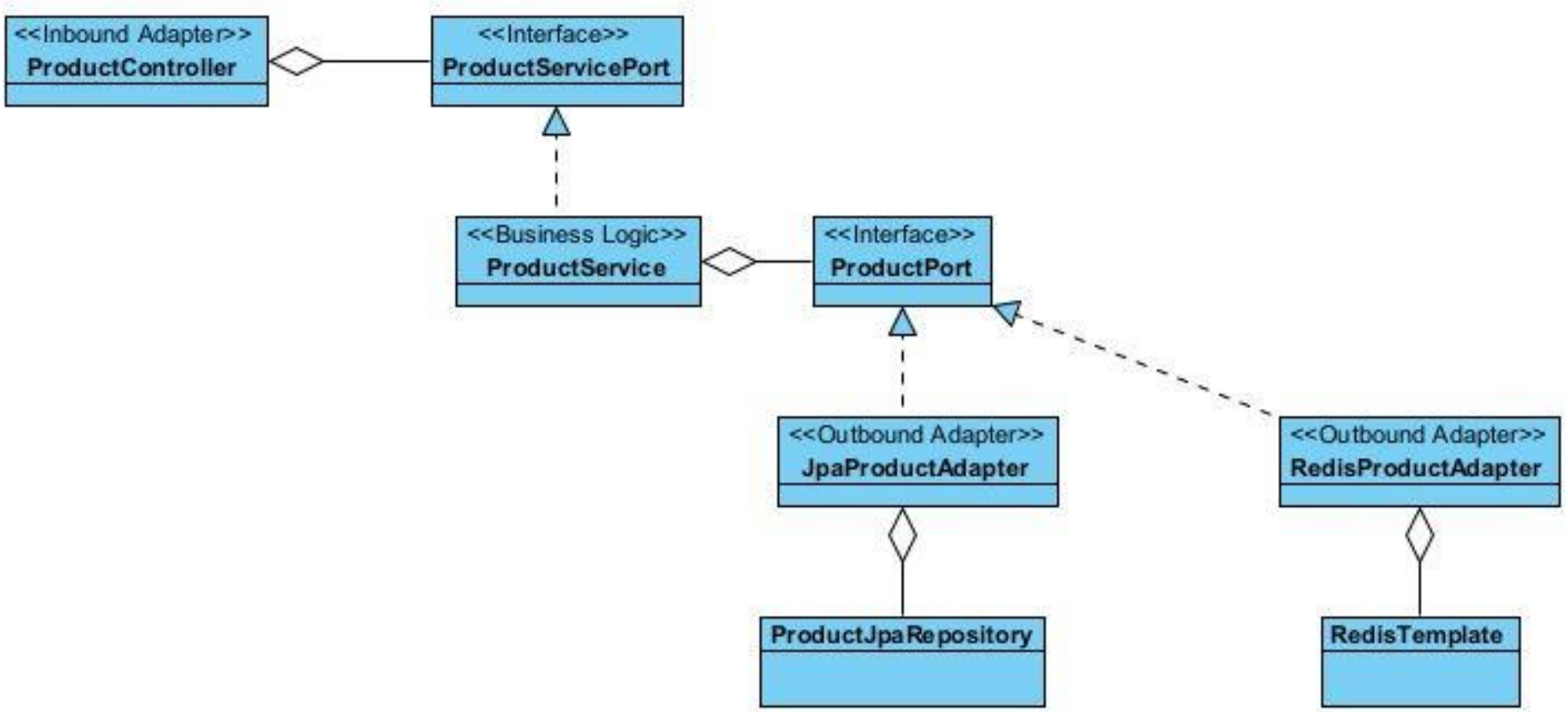
# Hexagonal Architecture (Ports and Adapters)

- The **hexagonal architecture**, introduced by Alistair Cockburn
- Architectural style that decouples an application's business logic (the core) from its external systems (e.g., databases, user interfaces, APIs).
- It achieves this by defining:
  - **Ports**: Interfaces that specify what the core needs or provides.
  - **Adapters**: Concrete implementations that connect ports to external systems.

# Hexagonal Architecture

- The application’s business logic (the “hexagon”) is at the center, surrounded by ports and adapters.
- External systems interact with the core only through these well-defined boundaries.
- **Ports**: Abstract interfaces that define:
  - **Inbound Ports**: Services the core provides to the outside (e.g., API endpoints).
  - **Outbound Ports**: Services the core requires from the outside (e.g., data persistence).
- **Adapters**: Concrete classes that implement ports, bridging the core to external systems (e.g., REST controllers, JPA repositories).

# Hexagonal Architecture



```java
package com.example.productservice.port;

import com.example.productservice.core.Product;

public interface ProductPort {
    Product findById(Long id);
    void save(Product product);
}
```

```java
package com.example.productservice.port;

import com.example.productservice.core.Product;

public interface ProductServicePort {
    Product updateStock(Long id, int quantity);
}
```

```java
package com.example.productservice.core;

import com.example.productservice.port.ProductPort;
import org.springframework.stereotype.Service;

@Service
public class ProductService implements ProductServicePort {
    private final ProductPort productPort;

    public ProductService(ProductPort productPort) {
        this.productPort = productPort;
    }

    @Override
    public Product updateStock(Long id, int quantity) {
        Product product = productPort.findById(id);
        product.reduceStock(quantity);
        productPort.save(product);
        return product;
    }
}
```

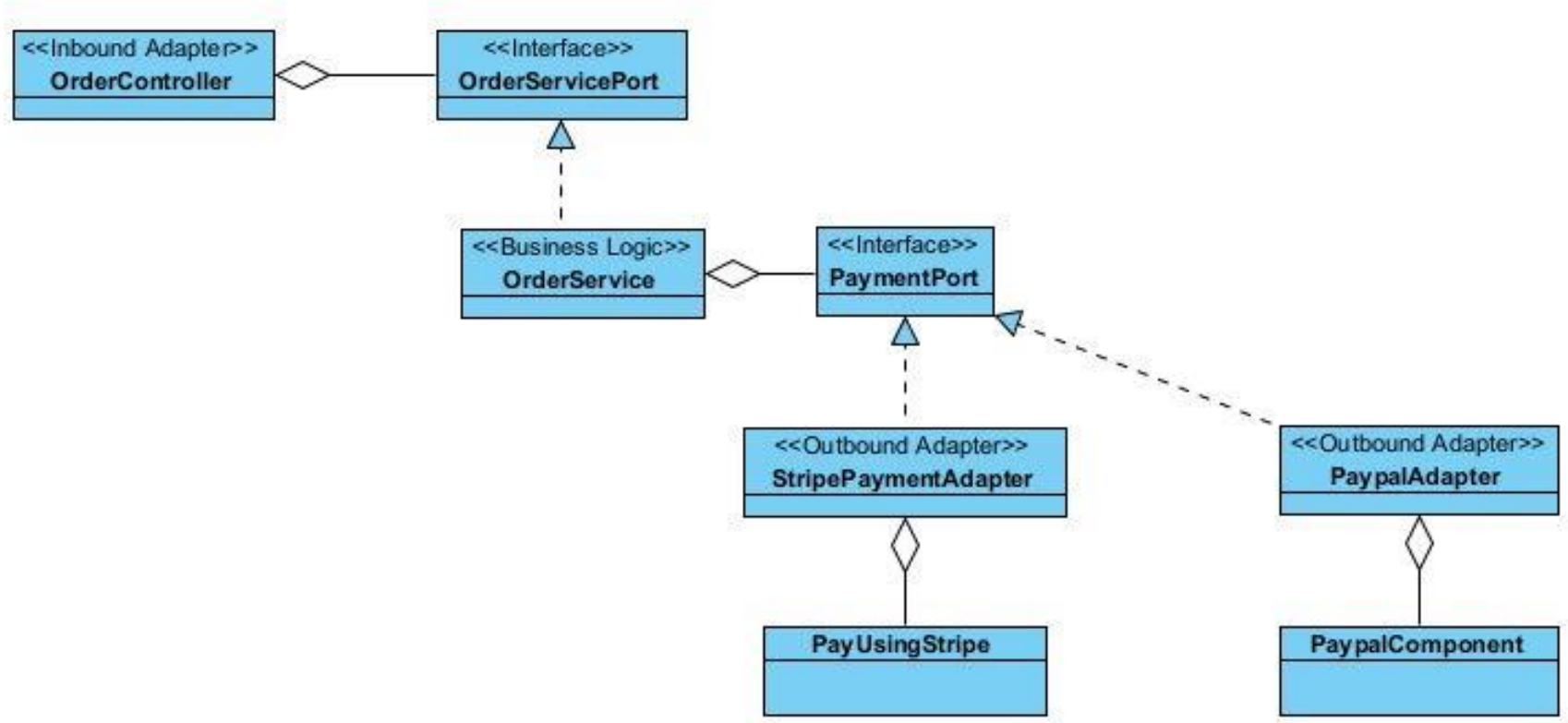
```java
import com.example.productservice.core.Product;
import com.example.productservice.port.ProductPort;
import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;

@Repository
public class JpaProductAdapter implements ProductPort {
    private final ProductJpaRepository repository;

    public JpaProductAdapter(ProductJpaRepository repository) {
        this.repository = repository;
    }

    @Override
    public Product findById(Long id) {
        return repository.findById(id)
            .orElseThrow(() -> new IllegalArgumentException("Product not found: " +
id));
    }

    @Override
    public void save(Product product) {
        repository.save(product);
    }
}

@Repository
interface ProductJpaRepository extends JpaRepository<Product, Long> {
}
```

```java
package com.example.productservice.adapter;

import com.example.productservice.core.Product;
import com.example.productservice.core.ProductService;
import org.springframework.web.bind.annotation.*;

@RestController
@RequestMapping("/products")
public class ProductController {
    private final ProductService productService;

    public ProductController(ProductService productService) {
        this.productService = productService;
    }

    @PutMapping("/{id}/stock")
    public Product updateStock(@PathVariable Long id, @RequestParam int quantity) {
        return productService.updateStock(id, quantity);
    }
}
```

# Providing Additional Implementation

```java
import com.example.productservice.core.Product;
import com.example.productservice.port.ProductPort;
import org.springframework.data.redis.core.RedisTemplate;
import org.springframework.stereotype.Component;

@Component
public class RedisProductAdapter implements ProductPort {
    private final RedisTemplate<String, Product> redisTemplate;

    public RedisProductAdapter(RedisTemplate<String, Product> redisTemplate) {
        this.redisTemplate = redisTemplate;
    }

    @Override
    public Product findById(Long id) {
        Product product = redisTemplate.opsForValue().get("product:" + id);
        if (product == null) {
            throw new IllegalArgumentException("Product not found: " + id);
        }
        return product;
    }

    @Override
    public void save(Product product) {
        redisTemplate.opsForValue().set("product:" + product.getId(), product);
    }
}
```

# Payment Example



# Benefits

- **Decoupling**: Business logic is independent of infrastructure (e.g., swap databases without changing the core).
- **Testability**: Mock adapters to test the core in isolation.
- **Flexibility**: Add new adapters (e.g., switch from JPA to MongoDB) without altering business logic.

# Final Words

- We have seen how CBSE improved software development
  - Modularity
  - Reusability
  - Flexibility
  - Maintainability
- We have explored the CBSE in Spring Boot’s way
  - Define components including Bean
  - IoC, DI
  - Flexibility of having multiple implementation
  - Use of middleware such as JPA and Spring Security
  - DDD and Hexagonal pattern